

KOUSTUBH PANDE

Full Stack Developer

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SUMMARY

Computer Science undergraduate and full stack developer with hands-on experience building and shipping production applications end-to-end, using JavaScript/TypeScript, React.js, Node.js, and Python across REST APIs and PostgreSQL-backed databases. Has built AI-driven and agentic tools that automate workflows, including a client-side retrieval system for codebase exploration and a real-time voice AI interview platform, using RAG pipelines and LLM APIs. Comfortable debugging, testing, and pushing features live in agile sprints and self-directed environments, with CI/CD pipelines already in place on personal and freelance projects. Delivered measurable outcomes such as a 25% revenue increase for a client e-commerce platform and 90+ Lighthouse scores across 40+ production UI components. Eager to take ownership of AI-driven products while continuing to build expertise in Ruby on Rails and Flutter alongside my existing full-stack experience.

TECHNICAL SKILLS

- **Languages:** JavaScript, TypeScript, Python, C++, Java
- **Frontend & MVC Frameworks:** React.js, Next.js, Redux, HTML5, CSS3, Tailwind CSS
- **Backend & APIs:** Node.js, Express.js, REST APIs, Flask, Django, Authentication (JWT, OAuth)
- **Databases:** PostgreSQL, MySQL, MongoDB
- **AI & Automation:** RAG, LangChain, LLM APIs, transformers.js, Generative AI, AI Agent workflows
- **DevOps & Version Control:** Git, GitHub, GitHub Actions, CI/CD Pipelines, Docker
- **Deployment:** AWS, Vercel, Render, Netlify
- **Currently Building Depth In:** Ruby on Rails, Flutter

EXPERIENCE

Founding Engineer / Co-founder | Singularity Works

Aug 2025 – Present

- Designed and built REST API architecture and integrations end-to-end within a 4-person founding team, applying efficient API design patterns and Docker-containerized services to keep CI/CD environments consistent
- Developed 40+ reusable, component-driven UI modules in React.js, applying code splitting and lazy loading to cut load time by 40 to 50% and reach 90+ Lighthouse scores
- Set up a monorepo-based architecture using pnpm, Bun, and TurboRepo, integrating CI/CD pipelines that cut deployment time by 30% while working with minimal direction in a fast-moving, agile startup

Tech Stack: React.js, Node.js, TypeScript, REST APIs, Docker, CI/CD, Git

Full Stack Developer | Freelance

Dec 2024 – Jun 2025

- Built and launched a full-stack e-commerce platform for a jewelry startup on the MERN stack, debugging and iterating through client feedback in agile sprints to drive a 25% revenue increase and a 20% lift in customer retention
- Shipped a role-based admin dashboard with JWT-based authentication, enabling real-time inventory and order management and cutting manual workload by 40%
- Extended a React and Redux web app with an Appwrite backend, raising user engagement by 30%, and delivered responsive front-ends with Next.js and Tailwind CSS for two additional clients

Tech Stack: JavaScript, React, Redux, Node.js, Express.js, MongoDB, JWT, Next.js

PROJECTS

Qode — AI-Powered Visual Codebase Exploration Tool | [Live Link](#)

Tech Stack: React, TypeScript, KuzuDB (WASM), Tree-sitter, Sigma.js, LangChain, transformers.js

- Built a client-side retrieval system that parses codebases into knowledge graphs using KuzuDB and Tree-sitter, giving an AI agent more accurate context and reducing hallucinated answers
- Set up GitHub Actions CI/CD with ruff, pyright, and pytest for automated linting, type-checking, and test coverage on every pull request

VerQ — AI-Powered Interview Practice Platform | [GitHub](#)

Tech Stack: React, Node.js, Express, MongoDB, Gemini API, Deepgram API, RAG

- Built a full-stack AI agent platform on the MERN stack with a RAG pipeline and real-time voice processing, automating personalized interview feedback with sub-300ms response latency
- Implemented JWT-based authentication and a low-latency Deepgram streaming pipeline paired with Gemini prompt chaining for real-time transcription and feedback

Bazaario — Multi-Vendor E-Commerce Platform | [GitHub](#)

Tech Stack: React.js, Node.js, Express.js, MongoDB, JWT, Chart.js

- Designed a multi-vendor marketplace with customizable storefronts, JWT-based RBAC authentication, and REST APIs for secure data flow, plus a Chart.js analytics dashboard that improved vendor operational efficiency by 15%

EDUCATION

B.Tech, Computer Science and Engineering — Vellore Institute of Technology

Sep 2023 – 2027

Bhopal, MP | CGPA: 8.4 / 10.0

CERTIFICATIONS AND ACTIVITIES

- Winner, HACKBYTE 4.0 — national-level 36-hour hackathon, IIITDM Jabalpur | [Link](#)
- Winner, 48-hour offline hackathon by GSoC Innovators Club, VIT Bhopal, powered by Unstop and InterviewBuddy
- Google IT Support Professional Certificate — issued by Google | [Link](#)
- Tech Team Lead, Mozilla Firefox Club and The Eureka Club